A Minimum-time Zero Vibration S-curve Command for an Overhead Crane with Actuator Limits

Ho Duc Tho, Naoki Uchiyama, and Kazuhiko Terashima

***Abstract*— S-curve commands are ubiquitous in industrial servo drives due to their simplicity and smoothness. Nevertheless, they need to be adjusted to adapt for flexible systems. This paper proposes a simple motion planning scheme using S-curve command for an overhead crane system, in which an issue of residual vibration suppression ought to be addressed. Specifically, the S-curve command gains the vibration suppression property by embedding essential conditions of a vibration-free transferring process. In order to enhance transportation productivity, minimum-time optimization problem will be resolved by parameterizing the S-curve command to establish constrained discrete nonlinear programs, in which maximum velocity and maximum acceleration of the actuator are explicitly taken into account. Global optimal solution is successfully obtained. It will be shown that, when infinite actuator limits are allowed, the proposed S-curve command is only *two* times slower than a discontinuous command formed by a step function and a zero vibration input shaper, not *four* times as suggested by the influential study [1]. Both simulation and experimental results are provided to verify effectiveness of the proposed method.**

I. Introduction

Overhead crane is widely utilized in various places such as factories, harbors, etc. to transfer heavy payloads point-to-point. The payload is suspended under the trolley and the trolley actuator relocates the payload within the workspace. Since the rope is flexible, the payload will exhibit residual vibration during and after the transportation. This undesirable oscillation significantly reduce the safety and productivity of the overhead crane system in practice. Consequently, two control objectives need to be realized: 1) driving the trolley to a desirable destination and 2) suppressing the swing movement of the payload. This is a challenging task since the overhead crane is an underactuated system—it has only one control input (trolley actuation) and two control outputs. Therefore, considerable interests were mounting on the vibration control for the overhead crane system. They can be classified into two main classes: feedback and feedforward controls. Those types can be combined to obtain a two degrees of freedom control system which could give a better control performance.

Feedback vibration controls for the overhead crane system are plentiful in the literature. Lee et al. [2] proposed a cascade control system, in which the position servo control and the anti-swing loop are designed based on the loop-shaping and the root locus technique respectively. Mori and Tagawa [3] applied a dual-model-matching controller while considering the limitation of horizontal acceleration. Feedback linearization and differential-flatness based methodology [4], which aims at transforming the nonlinear model to a linear one to utilize well-established linear controls, are also widely applied [5], [6]. The main disadvantage of the feedback linearization method lies in the fact that it is not robust to the parametric uncertainties and unmodeled dynamics. In order to deal with such circumstances, sliding-mode based controls [7], [8], adaptive controls [9], [10], energy and passivity-based nonlinear controls [11], [12] are preferable. To sum up, feedback control techniques showed their excellent properties in handling parametric uncertainties and external disturbances. However, they demand sophisticated and costly sensor structure which needs a regular maintenance; those are the reasons that feedback controllers are still not popular in practical cranes. In numerous contexts, feedforward controls are preferable in industry due to their simplicity, ease of implementation, as well as no additional measurement system is required.

Rest-to-rest maneuvers generated from the S-curve feedforward commands are widely used for flexible systems in general and for cranes in particular owning to their simplicity, versatility, and smoothness [13]–[15]. Most of the existing studies, nevertheless, omitted the actuator limits including maximum velocity and maximum acceleration in the design process; thus, the planned S-curve command may be unrealistic to apply on a practical system. Sun et al. [16] proposed a three-segment (i.e. bang-off-bang) acceleration profile and its modified versions for a two-dimensional overhead crane. A geometric approach based on the system phase portrait is employed to derive vibration suppression conditions for the S-curve profile. Hoang et al. [17] introduced a similar approach with [16] and extended that results by considering a family of acceleration profiles in stair form. Minimum-time solution, however, is not considered in the above-mentioned studies. In contrast to [16], [17] where the motion is planned in the acceleration level, the jerk-limited S-curve trajectory of [18] is established in the jerk level for an undamped flexible system. In [19], a discrete-time command profile is proposed using finite step segments which matches steps of the discrete command signal to that of the control hardware. Unfortunately, the system is forced to use both maximum velocity and maximum acceleration of the actuator, which possibly is an infeasible assumption for short distance maneuvers. In summary, to date, none of existing study solves the minimum transferring time problem for the Zero Vibration (ZV) S-curve command under restrictions imposed by actuator limitations.

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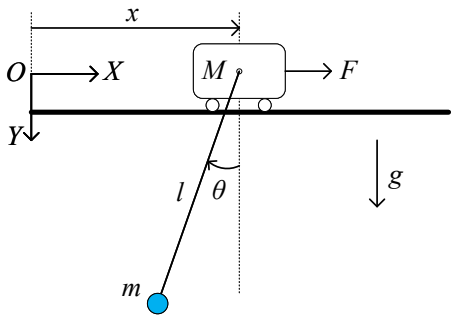

Fig. 1. Mathematical model of the overhead crane system.

In response to the previous discussions, this paper will establish minimum-time solution of the ZV S-curve position command for an overhead crane system considering both maximum velocity and maximum acceleration of the actuator. In this paper, the S-curve command is understood in the position level and it is created from a bang-off-bang acceleration trajectory. In order to realize a vibration-free transportation, the vibration suppression conditions at the end of maneuver are embedded to the S-curve command. Consequently, the resultant S-curve command—named S-curve–Vibration Suppression Embedded (SC–VSE)—has internal constraints on the duration of acceleration, uniform, and deceleration intervals, which strongly relate to the natural frequency of the system. It will be shown that conditions of the zero residual vibration are deficient, which leads to an under-determined issue. This point is then resolved together with the minimum transferring time problem by formulating constrained discrete nonlinear programs. The main challenge needs to be addressed in the paper is to obtain a global optimal solution of these optimization problems in closed form, with which the computational effort is low; thus, online command generation can be realized. In comparison with previous studies, contributions of this paper are listed as:

1) Actuator limits, including maximum velocity and maximum acceleration, are elaborately considered.
2) Global minimum-time solution of the SC–VSE is systematically formulated.
3) When infinite actuator limits are allowed, this paper will show that the SC–VSE is only *two* times slower than a discontinuous command formed by a step function and the ZV input shaper, not *four* times as suggested by the influential work in [1].

II. Problem formulation

Swinging motion of the payload resulted from the trolley movement is illustrated in Fig. 1, where $x(t)$ (m) and $\theta(t)$ (rad) are the trolley position and the swing angle respectively. The net force driving the trolley is denoted as $F(t)$. Furthermore, m (kg), M (kg), l (m), and g (= 9.81 m/s^2), in order, represents the payload mass, trolley mass, rope length, and gravitational acceleration. By utilizing the Euler-Lagrange method, dynamic of the overhead crane is described by:

$$(M+m)\,\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta = F \tag{1}$$

$$l\ddot{\theta} + g\sin\theta = \cos\theta\ddot{x} \tag{2}$$

From (1) and (2), it is recognized that one can either use $F(t)$ or $\ddot{x}(t)$ as the control input. For the motion planning task, $\ddot{x}(t)$ is the more appropriate choice due to the following reasons: i) the problematic frictional forces presented in $F(t)$ can be avoided, ii) the first and second integration of $\ddot{x}(t)$ will result in the velocity $\dot{x}(t)$ and the position $x(t)$ commands, which can be easily realized by ubiquitous servo drives, iii) the system dynamic is only dependent on the rope length l, which can be easily measured in practice. By using the small angle approximation, (2) can be linearized as:

$$l\ddot{\theta}(t) + g\theta(t) = \ddot{x}(t). \tag{3}$$

From (3), angular natural frequency ω_0 (rad/s) and natural period T (s) of the overhead crane system are given by $\omega_0 = \sqrt{g/l}$ and $T = 2\pi/\omega_0$. Note that the linear model (3) will be employed to design the SC–VSE in section III. Throughout the paper, it is assumed that the payload starts from rest, namely $x(0) = \dot{x}(0) = \theta(0) = \dot{\theta}(0) = 0$ [16]–[18]. Moreover, to realize a vibration-free transportation, the following terminal conditions must be fulfilled:

$$x(t_f) = x_r,\ \dot{x}(t_f) = 0,\ \theta(t_f) = 0,\ \dot{\theta}(t_f) = 0. \tag{4}$$

In (4), x_r (m) is the reference value of the trolley position. Without loss of generality, x_r is a positive constant, namely $x_r > 0$. The terminal time t_f will be minimized to enhance the transportation productivity (see section III).

In order to ensure the practical feasibility, the following maximum velocity and maximum acceleration of the trolley actuator must be strictly kept:

$$|\dot{x}(t)| \le v_{\max}, \quad |\ddot{x}(t)| \le a_{\max}, \tag{5}$$

where $v_{\max} > 0$ and $a_{\max} > 0$. The following section will be devoted to the formulation of the SC–VSE.

III. Design of the minimum-time SC–VSE

In this section, by embedding the conditions (4) of a vibration-free transportation, the SC–VSE can be formed. It will be later shown that the no-residual-swing conditions $\theta(t_f) = \dot{\theta}(t_f) = 0$ are deficient, which leads to an under-determined issue. This can be solved together with the minimum-time problem by establishing constrained discrete nonlinear programs. Simulation results are provided for several maneuvers with different values of $v_{\max}$ and $a_{\max}$ to verify the proposed minimum-time solution.

The baseline position S-curve trajectory is generated by integrating twice a bang-off-bang acceleration profile as depicted in Fig. 2. This piece-wise continuous function can be expressed by

$$\ddot{x}(t) = \begin{cases} a, & 0 \le t < t_1 \\ 0, & t_1 \le t < t_1 + t_2 \\ -a, & t_1 + t_2 \le t \le t_f\ (= t_1 + t_2 + t_3) \end{cases} \tag{6}$$

In (6), all parameters a, t_1, t_2, and t_3 are unknowns, whose pre-conditions are given by $0 < a \le a_{\max}$, $t_1 > 0$, $t_2 \ge 0$, and $t_3 > 0$. Furthermore, it is easy to see that the baseline S-curve obtained from (6) reaches its highest velocity (= at_1) at

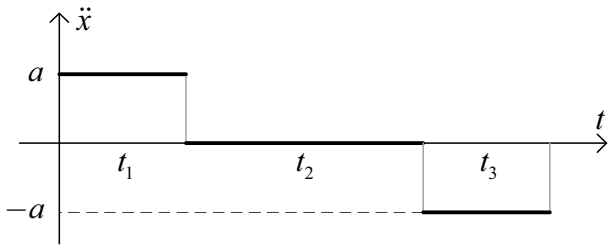

Fig. 2. Parameterized bang-off-bang acceleration profile of the trolley.

$t = t_1$. Therefore, the maximum velocity constraint $|\dot{x}(t)| \leq v_{\max}$ becomes $at_1 \leq v_{\max}$. The following subsection will establish additional constraints for these unknown variables to actively gain a vibration-free transportation.

A. Embedding the vibration-free transportation conditions

By exploiting the vibration-free transportation conditions (4), essential constraints on a, t_1, t_2, and t_3 can be further obtained. In the first step, response of the system (3) under the action of the control input (6) will be derived.

During the acceleration interval, namely $0 \leq t < t_1$, by substituting $\ddot{x}(t) = a$ into (3) and solving the second-order differential equation with a constant excitation, we have:

$$\begin{aligned} \theta(t) &= \frac{a}{g}\left(1 - \cos\omega_0 t\right), & \dot{\theta}(t) &= \frac{a\omega_0}{g}\sin\omega_0 t \\ x(t) &= 0.5at^2, & \dot{x}(t) &= at. \end{aligned} \tag{7}$$

Similarly, when $t_1 \leq t < t_1 + t_2$ (i.e. in the uniform interval), we have $\ddot{x}(t) = 0$, therefore

$$\begin{aligned} \theta(t) &= \frac{a}{g}\left[\cos\left(\omega_0\left(t - t_1\right)\right) - \cos\omega_0 t\right] \\ \dot{\theta}(t) &= -\frac{a\omega_0}{g}\left[\sin\left(\omega_0\left(t - t_1\right)\right) - \sin\omega_0 t\right] \\ x(t) &= at_1 t - 0.5at_1^2 \\ \dot{x}(t) &= at_1. \end{aligned} \tag{8}$$

Finally, when $t_1 + t_2 \leq t \leq t_f$ (or in the deceleration interval), we have $\ddot{x}(t) = -a$, hence:

$$\begin{aligned} \theta(t) &= \frac{a}{g}\left[\sum_{i=1}^{2}\cos\left(\omega_0\left(t - \sum_{j=1}^{i} t_j\right)\right) - \cos\omega_0 t - 1\right] \\ \dot{\theta}(t) &= \frac{-a\omega_0}{g}\left[\sum_{i=1}^{2}\sin\left(\omega_0\left(t - \sum_{j=1}^{i} t_j\right)\right) - \sin\omega_0 t\right] \\ x(t) &= -\frac{1}{2}at^2 + a\left(2t_1 + t_2\right)t - a\left(t_1^2 + t_1 t_2 + \frac{1}{2}t_2^2\right) \\ \dot{x}(t) &= a\left(-t + 2t_1 + t_2\right). \end{aligned} \tag{9}$$

From (9), the zero velocity condition of the trolley at t_f—$\dot{x}\left(t_f\right) = 0$—gives us $a\left(-t_f + 2t_1 + t_2\right) = 0$, which can be easily developed to $t_f - 2t_1 - t_2 = 0$ since $a > 0$. Note that $t_f = t_1 + t_2 + t_3$; thus, $t_1 = t_3$. This means that duration of the acceleration and deceleration periods must be similar. By using this result, the condition on the target trolley position, namely that $x\left(t_f\right) = x_r$, can be exploited as

$$at_1\left(t_1 + t_2\right) = x_r. \tag{10}$$

The condition of zero swing angle at t_f can be further developed by setting $\theta\left(t_f\right) = 0$ (in (9)). Hence, it yields:

$$\cos\left(\omega_0\left(t_1 + t_2\right)\right) + \cos\omega_0 t_1 = 1 + \cos\left(\omega_0\left(2t_1 + t_2\right)\right). \tag{11}$$

By using the cosine sum-to-product trigonometric identity $\cos p + \cos q = 2\cos\left(\left(p + q\right)/2\right)\cos\left(\left(p - q\right)/2\right)$ on the left-hand-side (LHS) and $\cos 2p = 2\cos^2 p - 1$ on the right-hand-side (RHS) of (11), we have

$$\cos\left(\frac{\omega_0\left(2t_1 + t_2\right)}{2}\right)\left[\cos\left(\frac{\omega_0 t_2}{2}\right) - \cos\left(\frac{\omega_0\left(2t_1 + t_2\right)}{2}\right)\right] = 0, \tag{12}$$

which results in

$$\left[\begin{aligned} &t_1 = k_1 T \\ &t_1 + t_2 = k_1 T \\ &2t_1 + t_2 = \left(k_1 - 0.5\right)T \end{aligned}\right., \quad k_1 \in \mathbb{N}^+. \tag{13}$$

Lastly, to ensure the zero-swing-rate condition $\dot{\theta}\left(t_f\right) = 0$, from (9), we need:

$$\sin\left(\omega_0\left(t_1 + t_2\right)\right) + \sin\omega_0 t_1 = \sin\left(\omega_0\left(2t_1 + t_2\right)\right). \tag{14}$$

The solution of (14) can be similarly obtained as

$$\left[\begin{aligned} &t_1 = k_2 T \\ &t_1 + t_2 = k_2 T \\ &2t_1 + t_2 = k_2 T \end{aligned}\right., \quad k_2 \in \mathbb{N}^+. \tag{15}$$

In summary, conditions of the zero residual vibration $\theta\left(t_f\right) = \dot{\theta}\left(t_f\right) = 0$ can be obtained by subsumed (13) and (15), which leads to either of the following case:

$$t_1 = k_1 T,\ k_1 \in \mathbb{N}^+ \tag{16}$$

$$\text{or } t_1 + t_2 = k_2 T,\ k_2 \in \mathbb{N}^+. \tag{17}$$

According to the above analysis, two equations $\theta\left(t_f\right) = 0$ and $\dot{\theta}\left(t_f\right) = 0$ are deficiently reduced down to a single equation—either (16) or (17). Therefore, we now have only two equations ((10)&(16) or (10)&(17)) to solve for three unknowns a, t_1, and t_2, which is clearly an under-determined problem. Furthermore, the actuator limits represented by $v_{\max}$ and $a_{\max}$ must be taken into account. Therefore, this paper will establish constrained discrete nonlinear programs to solve both under-determined and time-optimal problems while ensuring the actuator limits are strictly complied. The results of these optimization problems are nearly admitted in closed form; thus, no intensive computation is required. The following subsection will discuss on the first case (i.e. $t_1 = k_1 T,\ k_1 \in \mathbb{N}^+$) and Section III-C will be devoted to the second case (i.e. $t_1 + t_2 = k_2 T,\ k_2 \in \mathbb{N}^+$).

B. Case 1 of the SC–VSE: $t_1 = k_1 T,\ k_1 \in \mathbb{N}^+$

To begin, let us summarize essential equality and inequality constraints need to be taken into account: 1) trajectory constraints: $t_1 > 0$ and $t_2 \geq 0$, 2) actuator limits: $0 < a \leq a_{\max}$ and $at_1 \leq v_{\max}$ (see the beginning of Section III), 3) vibration-free transportation conditions:

$t_1 = k_1 T, (k_1 \in \mathbb{N}^+)$ and $a t_1 (t_1 + t_2) = x_r$ (see section III-A). By considering all of the above constraints and taking the total transferring time t_f as the cost function, the following optimization problem can be established:

$$\begin{aligned} \min_{a,t_1,t_2} \quad & t_f = 2t_1 + t_2 \\ \text{s.t.} \quad & a t_1 (t_1 + t_2) = x_r \\ & t_1 = k_1 T \, (k_1 \in \mathbb{N}^+), \; t_2 \geq 0 \\ & 0 < a \leq a_{\max}, \; a t_1 \leq v_{\max}. \end{aligned} \tag{18}$$

Note that (18) is a discrete nonlinear program since t_1 can only be a multiple of the natural period T, whereas a and t_2 are continuous. In (18), two variables t_1 and t_2 can be eliminated by using two equalities. Therefore, (18) becomes the following mixed nonlinear–integer program:

$$\begin{aligned} \min_{a,k_1} \quad & t_f = k_1 T + \frac{x_r}{a k_1 T} \\ \text{s.t.} \quad & k_1 \in \mathbb{N}^+ \\ & 0 < a \leq a_{\max}, \; a \leq \frac{x_r}{(k_1 T)^2}, \; a \leq \frac{v_{\max}}{k_1 T}. \end{aligned} \tag{19}$$

It will be later shown that, for a fixed k_1, (19) is actually a mixed quasi-linear–integer program, namely the original nonlinear optimization problem (19) can be converted to a linear one through a suitable variable substitution. In order to solve (19), a two-step procedure is proposed. Let us denote a set $\Delta(k_1)$ as

$$\Delta(k_1) = \left\{ k_1 T, \; \frac{x_r}{k_1 T a_{\max}}, \; \frac{x_r}{v_{\max}} \right\}. \tag{20}$$

In the first step, the optimal solution of (19) is established for each $k_1 \in \mathbb{N}^+$ by the following lemma.

Lemma 1. Consider the mixed nonlinear–integer optimization problem (19). For a fixed k_1 ($k_1 \in \mathbb{N}^+$), the minimizer $a^\dagger(k_1)$ and the optimum cost function $t_f^\dagger(k_1)$ of (19) are given by

$$a^\dagger(k_1) = \frac{x_r}{k_1 T \max \Delta(k_1)}; \; t_f^\dagger(k_1) = k_1 T + \max \Delta(k_1).$$

Proof. By assigning $z = x_r / (a k_1 T)$, the original nonlinear program (19) can be transformed to the following linear optimization problem:

$$\begin{aligned} \min_{z \in \mathbb{R}^+} \quad & t_f = k_1 T + z \\ \text{s.t.} \quad & z \geq k_1 T, \; z \geq \frac{x_r}{k_1 T a_{\max}}, \; z \geq \frac{x_r}{v_{\max}}. \end{aligned} \tag{21}$$

Note that the feasible region of (21) is $z \geq \max \Delta(k_1)$. It is easy to see that the minimizer of (21) is $z^\dagger = \max \Delta(k_1)$ and $t_f^\dagger = k_1 T + z^\dagger$. By using the relation $z = x_r / (a k_1 T)$, we have $a^\dagger(k_1) = x_r / (k_1 T \max \Delta(k_1))$ and $t_f^\dagger(k_1) = k_1 T + \max \Delta(k_1)$. □

In fact, through Lemma 1, the first step will provide all local minima of the optimization problem (19), which are infinite in number. In the second step, these local minima will be compared to each other to find the global optimum one. The result of the second step is summarized in Theorem 1.

Theorem 1. The global minimizer $k_1^\star$ and $a^\star$ of the mixed nonlinear–integer program (19) are:

$$k_1^\star = \begin{cases} 1 & \text{if } \bar{k}_1 = 1 \\ \bar{k}_1 & \text{if } \bar{k}_1 > 1, \; \mu < 0 \\ \bar{k}_1 - 1 & \text{if } \bar{k}_1 > 1, \; \mu \geq 0 \end{cases}; \; a^\star = \frac{x_r}{k_1^\star T \max \Delta(k_1^\star)},$$

in which $\mu = T + \max \Delta(\bar{k}_1) - \max \Delta(\bar{k}_1 - 1)$. The pivotal index is $\bar{k}_1 = 1$ if $\max \Delta(1) \neq x_r / (T a_{\max})$. Otherwise, namely $\max \Delta(1) = x_r / (T a_{\max})$, $\bar{k}_1$ is given by

$$\bar{k}_1 = \max \left\{ k_1 \in \mathbb{N}^+ : \max \Delta(k_1) = \frac{x_r}{k_1 T a_{\max}} \right\} + 1.$$

Moreover, the global minimum cost function is:

$$t_f^\star(k_1^\star) = k_1^\star T + \max \Delta(k_1^\star).$$

Proof. From the condition of the pivotal index $\bar{k}_1$, the proof is divided into two cases: $\max \Delta(1) \neq x_r / (T a_{\max})$ and $\max \Delta(1) = x_r / (T a_{\max})$, which will be detailed as follows. Note that the assumption $\max \Delta(1) = x_r / (T a_{\max})$ does not necessarily mean that k_1 cannot accept values other than 1.

1) Subcase 1: $\max \Delta(1) \neq x_r / (T a_{\max})$: Since $\max \Delta(1) \neq x_r / (T a_{\max})$, we can either have $\max \Delta(1) = T$ or $\max \Delta(1) = x_r / v_{\max}$. In the case that $\max \Delta(1) = T$, we have $T \geq x_r / v_{\max}$ and $T \geq x_r / (T a_{\max})$. Therefore, $x_r / v_{\max} \leq T < 2T < \ldots$ and $\ldots < x_r / (2 T a_{\max}) < x_r / (T a_{\max}) \leq T < 2T < \ldots$ Thus, $\max \Delta(k_1) = k_1 T$, which by Lemma 1 results in

$$t_f^\dagger(k_1) = k_1 T + \max \Delta(k_1) = k_1 T + k_1 T = 2 k_1 T. \tag{22}$$

Since $k_1 \in \mathbb{N}^+$, we have $t_f^\dagger(k_1 = 1) < t_f^\dagger(k_1 = 2) < t_f^\dagger(k_1 = 3) < \ldots$ Consequently, $k_1^\star = 1$.

For the remaining case, namely $\max \Delta(1) = x_r / v_{\max}$, we have $x_r / v_{\max} \geq x_r / (T a_{\max})$. As a result, $x_r / v_{\max} \geq x_r / (T a_{\max}) > x_r / (2 T a_{\max}) > \ldots$ Therefore, $\max \Delta(k_1) = \max \{ k_1 T, \; x_r / v_{\max} \}$ can be obtained. From that, for every $n \in \mathbb{N}^+$, we have

$$\begin{aligned} & t_f^\dagger(k_1 = 1 + n) - t_f^\dagger(k_1 = 1) \\ & = nT + \max \left\{ (1+n)T, \frac{x_r}{v_{\max}} \right\} - \frac{x_r}{v_{\max}}. \end{aligned} \tag{23}$$

From (23), if $\max \{ (1+n)T, \; x_r / v_{\max} \} = x_r / v_{\max}$, we have $t_f^\dagger(k_1 = 1 + n) - t_f^\dagger(k_1 = 1) = nT > 0$. Otherwise, namely if $\max \{ (1+n)T, \; x_r / v_{\max} \} = (1+n)T$, we have $t_f^\dagger(k_1 = 1+n) - t_f^\dagger(k_1 = 1) = nT + (1+n)T - x_r / v_{\max}$. But in this circumstance, $\max \{ (1+n)T, \; x_r / v_{\max} \} = (1+n)T$, thus $(1+n)T \geq x_r / v_{\max}$; therefore, $t_f^\dagger(k_1 = 1 + n) - t_f^\dagger(k_1 = 1) \geq nT > 0$. Consequently, it is concluded that when $\max \Delta(1) = x_r / v_{\max}$, we also have $k_1^\star = 1$. Finally, from the above arguments, it is summarized that $k_1^\star = 1$ when $\max \Delta(1) \neq x_r / (T a_{\max})$.

2) Subcase 2: $\max \Delta(1) = x_r / (T a_{\max})$: Flow of the proof for this case is briefly explained as follows. Firstly, it can be proven that the local minimum transferring time $t_f^\dagger(k_1)$ (see Lemma 1) is a strictly monotonically decreasing function with respect to k_1 when $k_1 \leq \bar{k}_1 - 1$. Secondly,

it is possible to prove that $t_f^\dagger(k_1)$ is strictly monotonically increasing for $\bar{k}_1 \leq k_1 < +\infty$. Finally, among $t_f^\dagger(k_1 = \bar{k}_1 - 1)$ and $t_f^\dagger(k_1 = \bar{k}_1)$, the smaller one is the global minimum solution. Details of the proof are omitted here due to lack of space. However, the results can be summarized as

$$\begin{cases} t_f^\dagger(k_1) < t_f^\dagger(k_1 - 1), & \forall k_1 \leq \bar{k}_1 - 1 \\ t_f^\dagger(k_1) < t_f^\dagger(k_1 + 1), & \forall k_1 \geq \bar{k}_1 \end{cases}, \; k_1 \in \mathbb{N}^+. \quad (24)$$

Therefore, the global optimal index of the mixed nonlinear–integer program (19) is the one among $\bar{k}_1 - 1$ and $\bar{k}_1$, which produces a smaller transferring time. By using the result in Lemma 1, we have

$$\begin{aligned} & t_f^\dagger(k_1 = \bar{k}_1) - t_f^\dagger(k_1 = \bar{k}_1 - 1) = \\ & T + \max \Delta(\bar{k}_1) - \max \Delta(\bar{k}_1 - 1). \end{aligned} \quad (25)$$

By using the definition of μ in Theorem 1 and (25), it is concluded that when $\mu \geq 0$, namely $t_f^\dagger(k_1 = \bar{k}_1) \geq t_f^\dagger(k_1 = \bar{k}_1 - 1)$, the global optimal index of (19) is $k_1^\star = \bar{k}_1 - 1$. Otherwise, namely $\mu < 0$, we have $k_1^\star = \bar{k}_1$. This completes the proof.

□

C. Case 2 of SC–VSE: $t_1 + t_2 = k_2 T,\ k_2 \in \mathbb{N}^+$

In order to draw a distinction with Case 1, in Case 2, the unknown parameters a, t_1, t_2, and t_f are denoted as $\tilde{a}$, $\tilde{t}_1$, $\tilde{t}_2$, and $\tilde{t}_f$ respectively. In a similar manner to Case 1, by gathering all essential trajectory, actuator limits, and vibration-free transportation constraints and taking the total transferring time as the cost function, the following discrete nonlinear program can be established:

$$\begin{aligned} \min_{\tilde{a}, \tilde{t}_1, \tilde{t}_2} \quad & \tilde{t}_f = 2\tilde{t}_1 + \tilde{t}_2 \\ \text{s.t.} \quad & \tilde{a}\tilde{t}_1 \left(\tilde{t}_1 + \tilde{t}_2\right) = x_r \\ & \tilde{t}_1 + \tilde{t}_2 = k_2 T \; (k_2 \in \mathbb{N}^+),\ \tilde{t}_1 > 0,\ \tilde{t}_2 \geq 0 \\ & 0 < \tilde{a} \leq a_{\max},\ \tilde{a}\tilde{t}_1 \leq v_{\max}. \end{aligned} \quad (26)$$

In (26), by using the equality constraints, two variables $\tilde{t}_1$ and $\tilde{t}_2$ can be eliminated; thus, it can transformed to the following mixed nonlinear–integer program:

$$\begin{aligned} \min_{k_2, \tilde{a}} \quad & \tilde{t}_f = k_2 T + \frac{x_r}{\tilde{a} k_2 T} \\ \text{s.t.} \quad & k_2 \geq \frac{x_r}{T v_{\max}},\ k_2 \geq \frac{1}{T}\sqrt{\frac{x_r}{a_{\max}}},\ k_2 \in \mathbb{N}^+ \\ & \frac{x_r}{(k_2 T)^2} \leq \tilde{a} \leq a_{\max}. \end{aligned} \quad (27)$$

By conducting an analogous procedure with Case 1, solution of the optimization problem (27) is stated in Theorem 2. To begin, let us denote a new set $\beta(k_2)$ as follows:

$$\beta(k_2) = \left\{ k_2,\ \frac{x_r}{T v_{\max}},\ \frac{1}{T}\sqrt{\frac{x_r}{a_{\max}}} \right\}. \quad (28)$$

Theorem 2. The global optimal solution of the mixed nonlinear–integer optimization problem (27) is:

$$\begin{aligned} & k_2^\star = \min\left\{ k_2 \in \mathbb{N}^+ : \max \beta(k_2) = k_2 \right\} \\ & \tilde{a}^\star = a_{\max}. \end{aligned}$$

Furthermore, the global minimum cost function is given by:

$$\tilde{t}_f^\star(k_2^\star) = k_2^\star T + \frac{x_r}{k_2^\star T a_{\max}}.$$

Proof. The first step of the proof is to find the local optimal solutions of (27) for each $k_2 \in \mathbb{N}^+$ satisfying $k_2 \geq \max\left\{ x_r/(T v_{\max}),\ (1/T)\sqrt{x_r/a_{\max}} \right\}$. In the second step, the local minima provided by the first step (which are infinite in number) will be compared to each other to justify the global optimum.

The first step can be done as follows. For a fixed $k_2 \in \mathbb{N}^+$ satisfying $k_2 \geq \max\left\{ x_r/(T v_{\max}),\ (1/T)\sqrt{x_r/a_{\max}} \right\}$, by introducing a new variable $\tilde{z} = x_r/(\tilde{a} k_2 T)$, (27) can be transformed to following simple linear optimization problem

$$\begin{aligned} \min_{\tilde{z} \in \mathbb{R}^+} \quad & \tilde{t}_f = k_2 T + \tilde{z} \\ \text{s.t.} \quad & \frac{x_r}{k_2 T a_{\max}} \leq \tilde{z} \leq k_2 T. \end{aligned} \quad (29)$$

It is easy to show that the minimizer $\tilde{z}^\dagger$ and the minimum cost function $\tilde{t}_f^\dagger$ of (29) are $\tilde{z}^\dagger(k_2) = x_r/(k_2 T a_{\max})$ and $\tilde{t}_f^\dagger(k_2) = k_2 T + \tilde{z}^\dagger(k_2)$. By using the relation $\tilde{z} = x_r/(\tilde{a} k_2 T)$, we have $\tilde{a}^\dagger(k_2) = a_{\max}$ and $\tilde{t}_f^\dagger = k_2 T + x_r/(k_2 T a_{\max})$.

Now, we can proceed the second step by solving the following optimization problem:

$$\begin{aligned} \min_{k_2 \in \mathbb{N}^+} \quad & \tilde{t}_f^\dagger(k_2) = k_2 T + \frac{x_r}{k_2 T a_{\max}} \\ \text{s.t.} \quad & k_2 \geq \max\left\{ \frac{x_r}{T v_{\max}},\ \frac{1}{T}\sqrt{\frac{x_r}{a_{\max}}} \right\}. \end{aligned} \quad (30)$$

Now, let us consider a function $\tilde{t}_f^\dagger(y)$ where $y \in \mathbb{R}^+$, $y \geq \max\left\{ x_r/(T v_{\max}),\ (1/T)\sqrt{x_r/a_{\max}} \right\}$. By taking the first derivative of $\tilde{t}_f^\dagger(y)$ with respect to y, it can be obtained that

$$\frac{d\tilde{t}_f^\dagger(y)}{dy} = T - \frac{x_r}{y^2 T a_{\max}}. \quad (31)$$

However, the feasible region of y gives us $y \geq (1/T)\sqrt{x_r/a_{\max}}$, thus $T \geq x_r/\left(y^2 T a_{\max}\right)$. By applying this result on (31), it can be obtained that $d\tilde{t}_f^\dagger(y)/dy \geq 0$, which means that the function $\tilde{t}_f^\dagger(y)$ is monotonically increasing with respect to y. By performing the substitution $y = k_2$, it can be also concluded that the function $\tilde{t}_f^\dagger(k_2)$ is monotonically increasing with respect to k_2. Therefore, the global optimal index $k_2^\star$ of (30) is the minimum integer satisfying $k_2^\star \geq \max\left\{ x_r/(T v_{\max}),\ (1/T)\sqrt{x_r/a_{\max}} \right\}$. In the set notation, the above result is equivalent to $k_2^\star = \min\left\{ k_2 \in \mathbb{N}^+ : \max \beta(k_2) = k_2 \right\}$. The proof is therefore completed. □

By using the results of Theorem 1 and Theorem 2, the global optimal solution of the SC–VSE in Case 1 and Case 2 can be computed.

D. Comparison between Case 1 and Case 2

Despite the fact that Case 1 ($t_1 = k_1 T$) and Case 2 ($t_1 + t_2 = k_2 T$) will both result in the vibration-free transferring process, they exhibit some fundamental distinct properties, which will be clarified in the following discussions. In Case 1, by substituting $t_1 = k_1 T$ ($k_1 \in \mathbb{N}^+$) into (8), we have

$$\theta(t) = \dot{\theta}(t) = 0, \ \forall t \in [t_1, t_1 + t_2). \tag{32}$$

The condition of (32) implies that, in Case 1, there is always no residual vibration during the uniform interval (i.e. $t_1 \leq t < t_1 + t_2$). In contrast, a similar property is not present in Case 2 except in the circumstance that the resultant $\tilde{t}_f^\star$ is also a multiple of the natural period T.

The second different property between Case 1 and Case 2 can be revealed when we let $v_{\max} \to +\infty$ and $a_{\max} \to +\infty$, namely infinite maximum velocity and maximum acceleration are allowed. In such context, for Case 1, we have

$$\lim_{\lambda \to +\infty} \max \left\{ T, \frac{x_r}{T a_{\max}}, \frac{x_r}{v_{\max}} \right\} = T, \tag{33}$$

where λ is defined by $\lambda = \min\{v_{\max}, a_{\max}\}$. Therefore, we have $\lim_{\lambda \to +\infty} \max \Delta(1) = T$. Consequently, according to Theorem 1, we have $\lim_{\lambda \to +\infty} k_1^\star = 1$. Hence, it yields

$$\lim_{\lambda \to +\infty} t_f^\star(k_1^\star) = T + T = 2T. \tag{34}$$

In a similar manner, for Case 2, it can be shown that $\lim_{\lambda \to +\infty} \max \beta(k_2) = k_2$. Therefore, we have $\lim_{\lambda \to +\infty} k_2^\star = 1$. As a result, based on Theorem 2, it can be obtained that

$$\lim_{\lambda \to +\infty} \tilde{t}_f^\star(k_2^\star) = T. \tag{35}$$

The results of (34) and (35) can be interpreted as follows: with infinite velocity and acceleration limits, the minimum transferring time of Case 1 is twice of the natural period T, whereas it only requires one natural period in Case 2.

The vibration-free transportation using SC–VSE contains two candidates: Case 1 and Case 2. Therefore, in practice, the SC–VSE scheme will choose the one which produces a smaller optimal transferring time. For instance, as $v_{\max} \to +\infty$ and $a_{\max} \to +\infty$, we have $t_f^\star(\to 2T) > \tilde{t}_f^\star(\to T)$; thus, Case 2 should be selected. Note that, in the influential work of [1], it was stated that the position S-curve (including both triangular and trapezoidal velocity profiles) must be *four* times longer than a command formed by shaping a step function with a ZV input shaper. According to the previous discussion, it has been shown that the above conclusion is not generally true, namely it is only correct for Case 1, but not for Case 2. This can be explained as follows. When a step position command is employed in [1], it automatically implies that infinite velocity and acceleration can be admitted. The shaping process between a step position command and the ZV input shaper, therefore, introduces a total transferring time of $T/2$. In the same condition, Case 2 of the SC–VSE is selected, namely a vibration-free transportation can be guaranteed with a total time of T (see (35)), which is only *two* times longer. The shortcoming of [1] stems from the fact that Case 2 is not fully considered in the analysis/design.

TABLE I

CONFIGURATION OF TESTCASES (TCS).

TC#	$v_{\max}$	$a_{\max}$	Rope length	Natural period
A1	0.25 m/s	0.05 m/s^2	$l = 1$ m	$T = 2.01$ s
A2	0.5 m/s	0.5 m/s^2	$l = 1.4$ m	$T = 2.37$ s

TABLE II

COMPUTATIONAL RESULTS OF THE MINIMUM-TIME SC–VSE.

TC#	Selection	$a^\star/\tilde{a}^\star$	$t_1^\star/\tilde{t}_1^\star$	$t_2^\star/\tilde{t}_2^\star$	$t_f^\star/\tilde{t}_f^\star$	$k_1^\star/k_2^\star$
A1	Case 1	0.05 m/s^2	4.01 s	0.97 s	8.99 s	$k_1^\star = 2$
A2	Case 2	0.5 m/s^2	0.84 s	1.53 s	3.21 s	$k_2^\star = 1$

Case 1: Section III-B, Case 2: Section III-C.

IV. SIMULATION AND EXPERIMENTAL RESULTS

The proposed motion planning scheme now will be implemented on a real overhead crane system, which is illustrated in Fig. 3. The actuator is a YASKAWA USAFED-09C2L1K AC servo motor operating in the position control mode. MATLAB/Simulink is utilized to compose the program and build the executing file via Real Time Workshop. A Linux PC will operate the executing file in real time. Moreover, a pair of laser sensors KEYENCE VG-035 is used to measure the swing angle.

Two testcases A1–A2 (see Table I) with different values of $v_{\max}$ and $a_{\max}$ are employed to compute the minimum-time SC–VSE. Following to the discussion in Section III-D, among Case 1 and Case 2, we would choose the one which produces a smaller transferring time. Therefore, according to the computational results in Tables II, Case 1 is chosen for testcase A1 whereas testcase A2 uses Case 2. The average computational time for two testcases is 0.009 second on a Core i5 personal computer with 8GB of RAM.

The proposed scheme now will be compared with the related study of [16]. Note that, to draw a fair comparison, the constraint on the maximum allowable angle in [16] is relaxed (i.e. $\theta_{ub} = +\infty$). In a similar family of bang-off-bang acceleration profiles, the proposed minimum-time S-

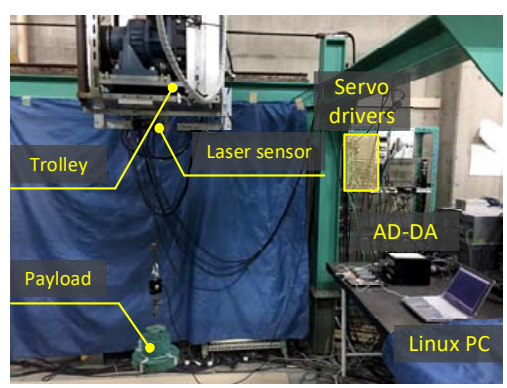

Fig. 3. Structure of the experimental overhead crane system.

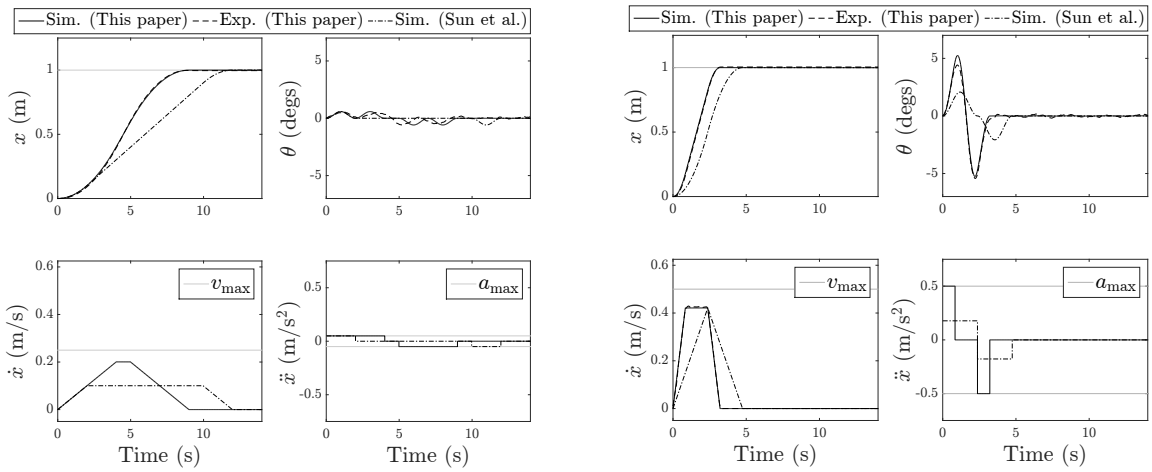

Fig. 4. Experimental results and comparisons with Sun et al. [16] in testcase A1 (left) and testcase A2 (right). Simulation (Sim.) and experimental (Exp.) results are in complete agreement.

curve command is 2.98 seconds and 1.53 seconds faster than [16] in testcase A1 and testcase A2 respectively (see Fig. 4).

Fig. 4 also depicts the experimental results of the minimum-time SC–VSE. In all testcases, it can be seen that the trolley precisely reaches the desirable destination while the vibration is nearly suppressed. Moreover, the maximum velocity and the maximum acceleration constraints of the actuator are strictly complied. The experimental results totally agree with the simulation results, which verifies the correctness of the entire modeling and control design process. Therefore, effectiveness of the proposed method is strongly confirmed.

V. Conclusions and Future Work

In this paper, a simple motion planning scheme using S-curve position command is proposed for an overhead crane to obtain a vibration-free transportation. Both maximum velocity and maximum acceleration of the actuator are elaborately taken into account. It has been shown that the zero residual vibration conditions at the end of transportation are deficient, which leads to the under-determined issue. This matter is then resolved together with the minimum transferring time problem by solving constrained discrete nonlinear programs. The global optimal solution is admitted in closed form; thus, no intensive computation is required. Both simulation and experimental results are provided to verify the effectiveness of the proposed method and they are in a complete agreement. Our future work is to extend the results of this paper to a three-dimensional overhead crane. Moreover, robustness of the proposed scheme with respect to parametric uncertainties will be analyzed.

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